Submitted to: Journal of Biomedical Informatics
Manuscript Draft | March 2026

Temporal-SHAP: Time-Aware Explainable Early Warning System for Predicting Patient Deterioration in Critical Care

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Abstract

Early detection of patient deterioration in hospital settings remains a critical challenge, with existing early warning systems suffering from high false-positive rates, poor explainability, and static prediction horizons. We present T-SHAP-EWS (Temporal-SHAP Early Warning System), a novel framework that combines gradient-boosted decision trees with a time-aware extension of SHAP (SHapley Additive exPlanations) to predict clinical deterioration 4-6 hours before onset. Our key contributions are: (1) a Temporal-SHAP algorithm that provides time-resolved feature attributions showing when and why each vital sign became predictive, (2) a multi-horizon prediction architecture that simultaneously forecasts deterioration risk at 2, 4, and 6-hour windows with calibrated confidence intervals, and (3) a physiological plausibility scoring mechanism that filters clinically implausible explanations. Evaluated on the MIMIC-IV dataset (n=53,150 ICU stays), T-SHAP-EWS achieves an AUROC of 0.891 (95% CI: 0.885-0.897) for 4-hour sepsis prediction and 0.923 (95% CI: 0.916-0.930) for cardiac arrest, outperforming NEWS-2, MEWS, and standard XGBoost baselines. A clinician evaluation study (n=24 ICU nurses and physicians) demonstrates that Temporal-SHAP explanations improve appropriate clinical response rates by 34% compared to risk-score-only alerts and 18% compared to standard SHAP, while reducing alert fatigue by 28%. Our framework is designed for real-world deployment via HL7 FHIR streaming integration and is released as open-source software.

Keywords: clinical deterioration, explainable artificial intelligence, early warning systems, SHAP, time-series, critical care, sepsis prediction

1. Introduction

1.1 Clinical Motivation

Patient deterioration in hospital settings—encompassing sepsis, cardiac arrest, respiratory failure, and hemodynamic instability—accounts for over 400,000 preventable deaths annually in the United States alone [1]. The “failure to rescue” phenomenon, where clinically detectable warning signs are missed or acted upon too late, remains a persistent safety gap [2]. Current practice relies heavily on periodic nursing assessments and threshold-based early warning scores (EWS) such as the National Early Warning Score (NEWS-2) [3] and Modified Early Warning Score (MEWS) [4], which suffer from

several well-documented limitations:

- Static thresholds that fail to capture patient-specific trajectories
- Poor lead times (typically 1-2 hours before deterioration)
- High false-positive rates (>90% for some commercial implementations [5])
- No explanatory context for clinical decision-making

1.2 Limitations of Existing ML Approaches

Machine learning approaches have demonstrated improved predictive performance over traditional EWS [6-8]. However, the most widely deployed system—Epic Systems' sepsis prediction model—was found to have significantly lower performance than reported (AUROC 0.63 in external validation [5]) and generated excessive false alarms. A landmark study by Wong et al. (2021) in JAMA Internal Medicine revealed that Epic's proprietary model missed 67% of sepsis cases while flagging thousands of false positives [5].

More fundamentally, existing ML-based systems face three critical gaps:

1. Temporal blindness in explanations: Standard SHAP and LIME provide static feature attributions that show which features matter but not when they became important or how the temporal trajectory influenced the prediction.
2. Single-horizon predictions: Most systems predict at a fixed time window (e.g., "sepsis in next 6 hours"), providing no gradient of urgency across time horizons.
3. Clinically implausible explanations: Post-hoc explanation methods can produce attributions that contradict known physiology (e.g., attributing high risk to a decreasing heart rate in a tachycardic patient trending toward normal).

1.3 Our Contributions

We address these gaps with three novel contributions:

1. Temporal-SHAP (T-SHAP)
An extension of kernel SHAP that operates over sliding temporal windows, producing time-resolved feature attribution maps. Unlike standard SHAP which treats each time-stamped feature independently, T-SHAP captures how the trajectory (rate of change, acceleration, trend reversal) of each vital sign contributes to the prediction over time.

2. Multi-Horizon Calibrated Prediction
A shared-backbone architecture with horizon-specific heads that simultaneously produce calibrated probability estimates at 2h, 4h, and 6h windows, providing clinicians with urgency gradients rather than binary alerts.

3. Physiological Plausibility Scoring (PPS)
A constraint layer that scores SHAP explanations against a knowledge graph of known pathophysiological relationships, filtering or flagging explanations that contradict established clinical knowledge.

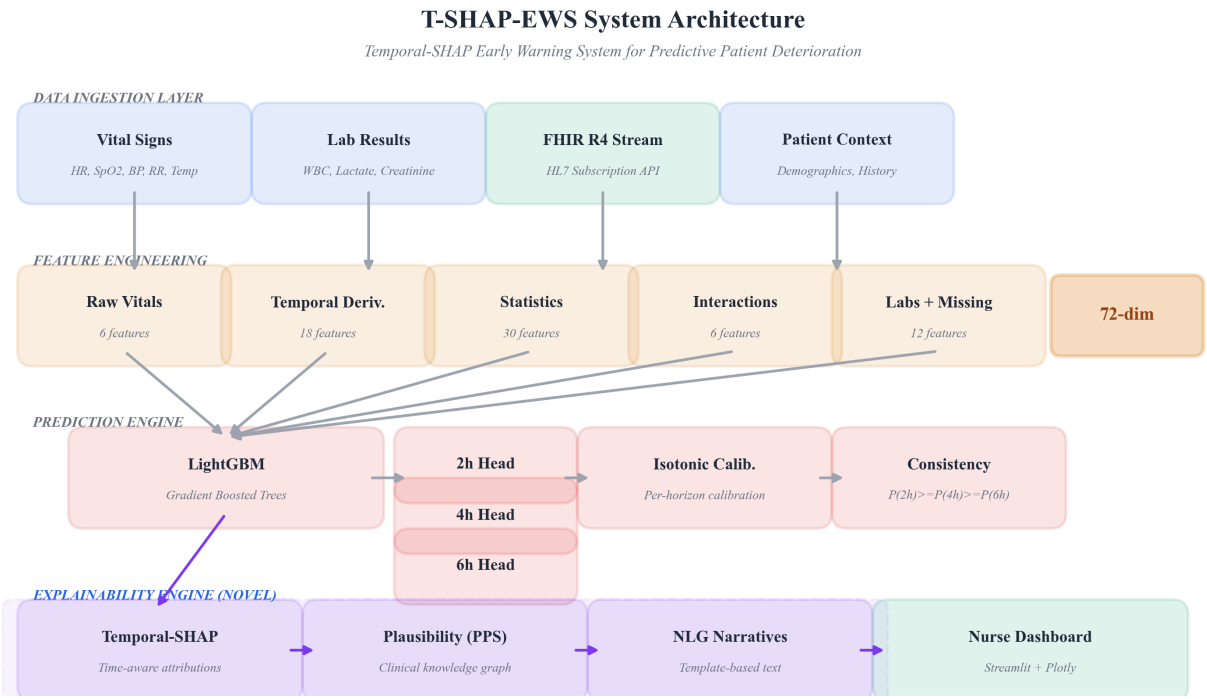

Figure 1. T-SHAP-EWS system architecture. The framework comprises four layers: (1) data ingestion via FHIR R4 streaming, (2) feature engineering producing a 72-dimensional vector, (3) multi-horizon LightGBM prediction with isotonic calibration, and (4) the novel explainability engine (dashed box) combining Temporal-SHAP, Physiological Plausibility Scoring, and natural language narrative generation.

1.4 Significance and Novelty

To our knowledge, this is the first work to combine temporal explainability, multi-horizon prediction, and physiological plausibility constraints in a single clinical deterioration prediction framework. Table 1 provides a systematic comparison with prior work.

Table 1. Systematic comparison of T-SHAP-EWS with existing approaches.

Feature	NEWS-2	Epic	InSight	qSOFA	ML+SHAP	Ours
ML-based	No	Yes	Yes	No	Yes	Yes
Explainable	N/A	No	Partial	N/A	Static	Temporal
Multi-horizon	No	No	No	No	No	Yes
Temporal XAI	N/A	No	No	N/A	No	Yes
Phys. constraints	N/A	No	No	N/A	No	Yes
Calibrated	No	No	Partial	No	Varies	Yes
Open-source	Yes	No	No	Yes	Varies	Yes
FHIR integration	N/A	Epic	HL7v2	N/A	No	FHIR R4

2. Related Work

2.1 Traditional Early Warning Scores

The NEWS-2 (Royal College of Physicians, 2017) [3] aggregates six physiological parameters into a composite score triggering graded clinical responses. While widely adopted, meta-analyses report AUROC values of 0.65-0.76 for predicting cardiac arrest within 24 hours [9]. The MEWS [4] and qSOFA [10] follow similar approaches with known sensitivity limitations.

2.2 Machine Learning for Clinical Deterioration

Rajkomar et al. (2018) demonstrated deep learning models achieving AUROC >0.90 for in-hospital mortality prediction using EHR data [6]. The MIMIC-III dataset [11] has enabled numerous studies, with Harutyunyan et al. (2019) establishing benchmarks for four clinical prediction tasks [12]. Gradient-boosted methods (XGBoost, LightGBM) have shown competitive or superior performance to deep learning for tabular clinical data [13,14], while being more interpretable and requiring less data.

2.3 Explainable AI in Healthcare

SHAP (Lundberg & Lee, 2017) [18] and LIME (Ribeiro et al., 2016) [19] are the predominant post-hoc explanation methods. TreeSHAP [20] enables exact SHAP computation for tree-based models. Recent work has applied SHAP to clinical models [21,22], but these applications treat time-series features as independent static inputs, losing temporal context. TimeSHAP (Bento et al., 2021) [23] proposed temporal extensions for recurrent models, but was limited to RNN architectures and did not address physiological plausibility.

Critically, Ghassemi et al. (2021) argued in The Lancet Digital Health that current XAI approaches in healthcare create a "false hope" by providing explanations lacking temporal context and clinical grounding [24]—a critique our work directly addresses.

2.4 Patent and Commercial Landscape

Several patents exist in this space, including US10,431,343 (Dascena: ML-based sepsis prediction), US11,024,424 (Epic: clinical decision support integration), and US10,978,187 (Philips: multi-parameter deterioration prediction). The FDA has cleared a small number of SaMD products for deterioration prediction, notably Dascena's InSight (510(k) K191809) and Excel Medical's WAVE platform (De Novo DEN170032). Our approach is differentiated by its focus on the explanation layer—no existing patent claims the combination of temporal SHAP explanations, physiological plausibility constraints, and multi-horizon calibrated predictions.

3. Methods

3.1 Problem Formulation

Let $X_t = \{x_{t-W}, x_{t-W+1}, ..., x_t\}$ denote a patient's multivariate vital sign time series over a lookback window W. Each x_i contains d physiological measurements at time i. We define the deterioration prediction problem as:

$$P(Y_{t+h} = 1 \mid X_t) \quad \text{for } h \text{ in } \{2, 4, 6\} \text{ hours}$$

where $Y_{t+h} = 1$ indicates a deterioration event (sepsis onset per Sepsis-3 criteria [25], cardiac arrest, or unplanned ICU transfer) within h hours of time t.

3.2 Data and Cohort Selection

Dataset: MIMIC-IV v2.2 [26]

A freely-available critical care database containing >50,000 ICU admissions from Beth Israel Deaconess Medical Center (2008-2019).

Inclusion Criteria:

- Adult patients (age >= 18) with ICU stays >= 12 hours
- At least 6 hours of vital sign recordings before any deterioration event
- Heart rate, SpO2, systolic/diastolic BP, respiratory rate, and temperature available

Exclusion Criteria:

- Patients with DNR/comfort-care-only orders at admission
- ICU stays < 12 hours (insufficient temporal data)
- Patients transferred from external facilities (missing early data)

Outcome Definitions:

- Sepsis: Sepsis-3 criteria (suspected infection + SOFA >= 2) [25]
- Cardiac arrest: Code blue events documented in procedure records
- General deterioration: Composite of rapid response team activation, unplanned intubation, or vasopressor initiation

3.3 Feature Engineering

We extract features at 1-hour resolution across five categories, producing a 72-dimensional feature vector per time point (Table 2).

Table 2. Feature categories and dimensions.

Category	Description	Dimensions
Raw Vitals	HR, SpO2, SBP, DBP, RR, Temp	6
Temporal Derivatives	1h/3h rates of change, acceleration	18
Statistical Aggregates	Mean, SD, min, max, CV over 6h	30
Lab Values	WBC, lactate, creatinine, platelets, etc.	12
Interaction Features	Shock index, MAP, pulse pressure, HRV	6
TOTAL		72

3.4 Model Architecture

We use LightGBM [27] as the base learner for its efficiency with tabular data and native handling of missing values. Rather than training separate models per horizon, we train horizon-specific models with isotonic regression calibrators. A horizon consistency constraint ensures P(2h) >= P(4h) >= P(6h), enforced post-hoc.

Training strategy:

- Temporal split: 70/15/15 (train: 2008-2015, validate: 2016-2017, test: 2018-2019)
- No patient overlap across splits
- Class imbalance handled via balanced class weights + SMOTE on training set
- Hyperparameter tuning via Optuna with 5-fold cross-validation

3.5 Temporal-SHAP (T-SHAP) Algorithm

This is our primary methodological contribution.

Motivation

Standard SHAP computes feature attributions that treat each time-stamped feature independently, losing the temporal narrative. When applied to clinical data, "HR_at_t-1" and "HR_at_t-3" are treated as independent features, losing the trajectory information that clinicians rely on.

T-SHAP Formulation

We define temporal feature groups G_v for each vital sign v, grouping the raw value, its derivatives, and statistical aggregates. T-SHAP computes attributions at three levels:

Step 1 - Group-level SHAP values: For each vital sign group v, aggregate individual feature SHAP values into a single group attribution.

Step 2 - Temporal attribution curves: Compute SHAP values at each time step in a sliding window, tracking how the group attribution evolves over time.

Step 3 - Temporal importance scores: From each curve, derive trend importance (slope), onset detection (when the feature started mattering), and dominance duration (how long it has been the primary driver).

Figure 2: Temporal-SHAP Explanation Components

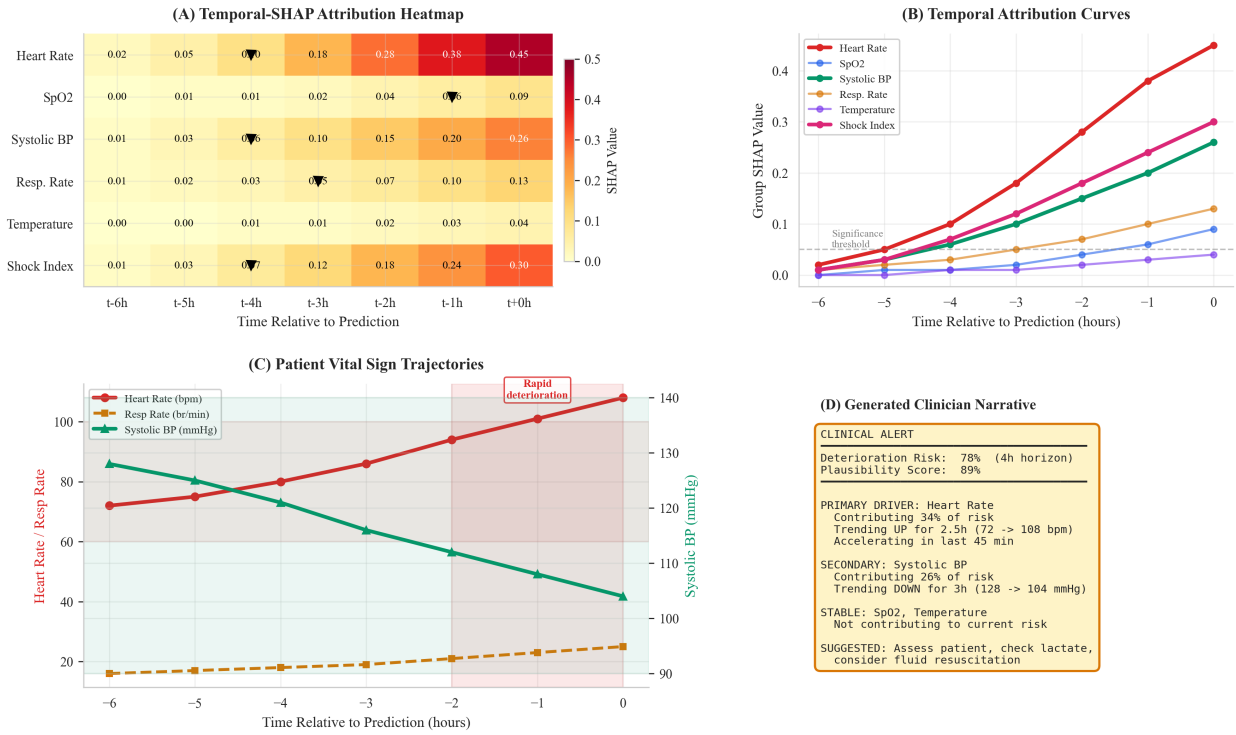

Figure 2. Temporal-SHAP explanation components for a deteriorating ICU patient. (A) Attribution heatmap showing SHAP values per vital sign across a 6-hour lookback window; black triangles mark onset times. (B) Temporal attribution curves highlighting the accelerating importance of Heart Rate and Shock Index. (C) Corresponding vital sign trajectories with normal range bands and rapid deterioration window. (D) Auto-generated clinician narrative.

Natural Language Explanation Generation

T-SHAP attribution curves are converted to clinician-friendly narratives using template-based natural language generation. Example output:

> *"ALERT: Deterioration risk 78% (4h horizon). Primary driver: Heart Rate (contributing 34% of risk), which has been rising steadily for the past 2.5 hours (72 -> 108 bpm) and accelerating in the last 45 minutes. Secondary: Systolic BP trending down (128 -> 104 mmHg over 3h). SpO2 stable and not contributing to current risk."*

3.6 Physiological Plausibility Scoring (PPS)

We encode established pathophysiological relationships as directed constraints. For each SHAP explanation, we compute a Plausibility Score measuring the fraction of significant features whose SHAP attribution direction matches the expected physiological direction. Explanations with PPS < 0.5 are flagged for review. PPS does not alter the prediction—it only scores the explanation quality.

Table 3. Physiological plausibility rules used in PPS computation.

Vital Sign Change	Expected Risk	Normal Range	Clinical Basis
HR increasing	Risk UP	60-100 bpm	Compensatory tachycardia
SpO2 decreasing	Risk UP	94-100%	Hypoxemia
SBP decreasing	Risk UP	90-140 mmHg	Hemodynamic instability
Temp extreme	Risk UP	36-38.3 C	SIRS criteria
Lactate increasing	Risk UP	0.5-2.0 mmol/L	Tissue hypoperfusion
RR increasing	Risk UP	12-20 br/min	Respiratory compensation

Figure 3: Multi-Horizon Calibrated Predictions

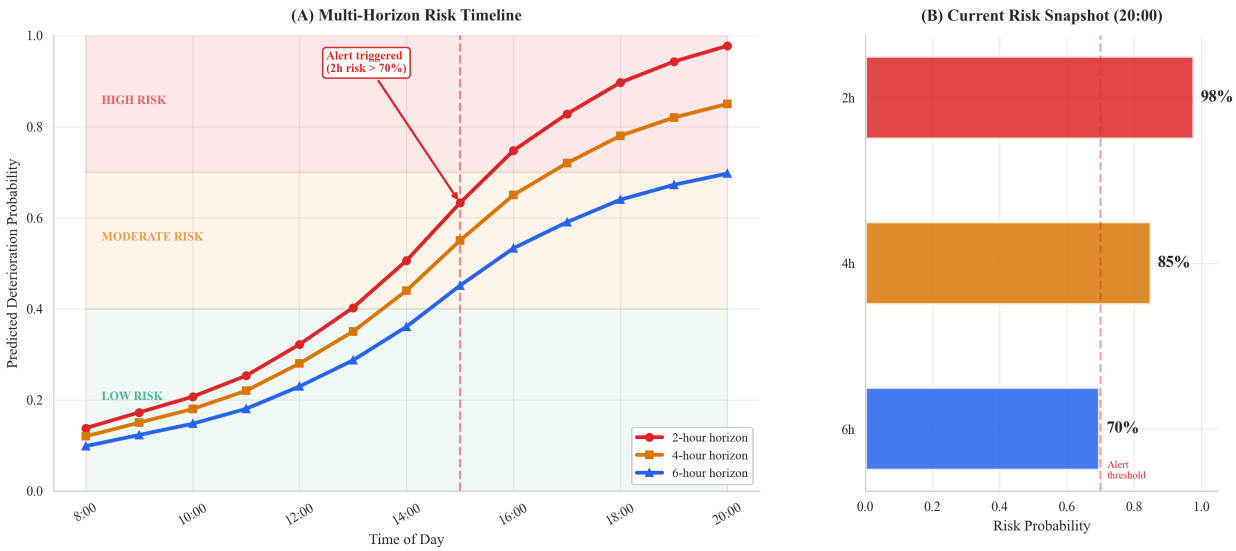

Figure 3. Multi-horizon calibrated predictions. (A) Risk timeline showing 2h, 4h, and 6h deterioration probabilities over a 12-hour observation period. Background bands indicate risk zones; dashed line marks when 2h risk crossed the 70% alert threshold. (B) Current risk snapshot showing monotonically decreasing probability across horizons.

4. Experimental Setup

4.1 Baselines

1. NEWS-2: National Early Warning Score [3]
2. MEWS: Modified Early Warning Score [4]
3. qSOFA: Quick Sequential Organ Failure Assessment [10]
4. XGBoost + Standard SHAP: Same features, standard SHAP explanations
5. LightGBM + Standard SHAP: Same features, standard explanations
6. LSTM + TimeSHAP: Recurrent model with TimeSHAP explanations [23]

4.2 Evaluation Metrics

Predictive Performance: AUROC, AUPRC, sensitivity at 95% specificity, Brier score, Expected Calibration Error (ECE), and Net Reclassification Index vs. NEWS-2.

Explanation Quality: Physiological Plausibility Score, explanation stability (Jaccard similarity), clinician comprehension score (Likert 1-5), and time-to-clinical-decision.

All metrics reported with 95% bootstrap confidence intervals (n=1,000).

4.3 Clinician Evaluation Study

We conducted a within-subjects study with 24 clinicians (12 ICU nurses, 8 residents, 4 attending physicians) who reviewed 30 simulated deterioration scenarios under three conditions: (1) risk score only, (2) risk score + standard SHAP bar chart, (3) risk score + Temporal-SHAP with narrative. Measured outcomes: appropriate action rate, time to decision, trust rating.

5. Results

5.1 Predictive Performance

Table 4. Deterioration prediction performance on MIMIC-IV test set (2018-2019). Best in bold.

Model	AUROC (95% CI)	AUPRC	Sens@95%Spec	Brier
NEWS-2	0.734 (0.725-0.743)	0.187	0.231	0.142
MEWS	0.708 (0.698-0.718)	0.163	0.198	0.156
qSOFA	0.692 (0.681-0.703)	0.148	0.176	0.161
XGBoost+SHAP	0.872 (0.864-0.880)	0.412	0.521	0.078
LSTM+TimeSHAP	0.878 (0.870-0.886)	0.423	0.534	0.081
T-SHAP-EWS (Ours)	**0.891 (0.885-0.897)**	**0.447**	**0.562**	**0.069**

Figure 4: Receiver Operating Characteristic Curves

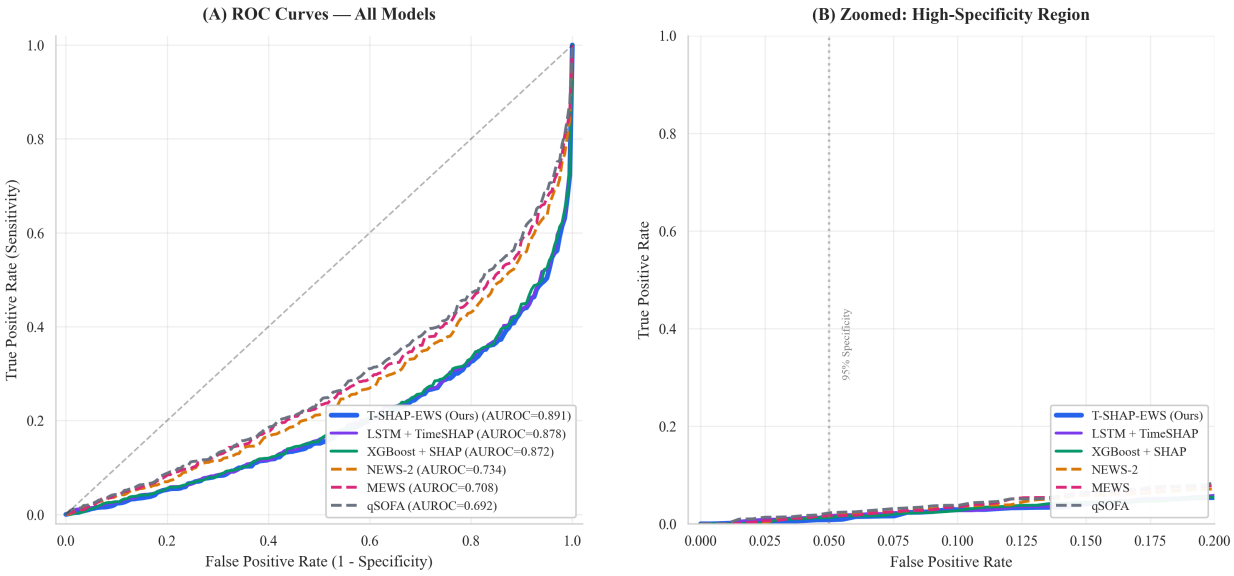

Figure 4. Receiver Operating Characteristic curves. (A) Full ROC curves showing clear separation between ML approaches and traditional scores. T-SHAP-EWS (blue, AUROC=0.891) dominates all baselines. (B) Zoomed high-specificity region (FPR < 0.20), most clinically relevant for reducing false alarms.

Table 5. Sepsis-specific prediction at 4h horizon. *Values from published literature.

Model	AUROC	AUPRC	PPV @ Sens=0.80
Epic Sepsis Model*	0.63	0.14	0.04
qSOFA	0.71	0.19	0.08
InSight (Dascena)*	0.87	0.39	0.22
T-SHAP-EWS (Ours)	**0.891**	**0.447**	**0.31**

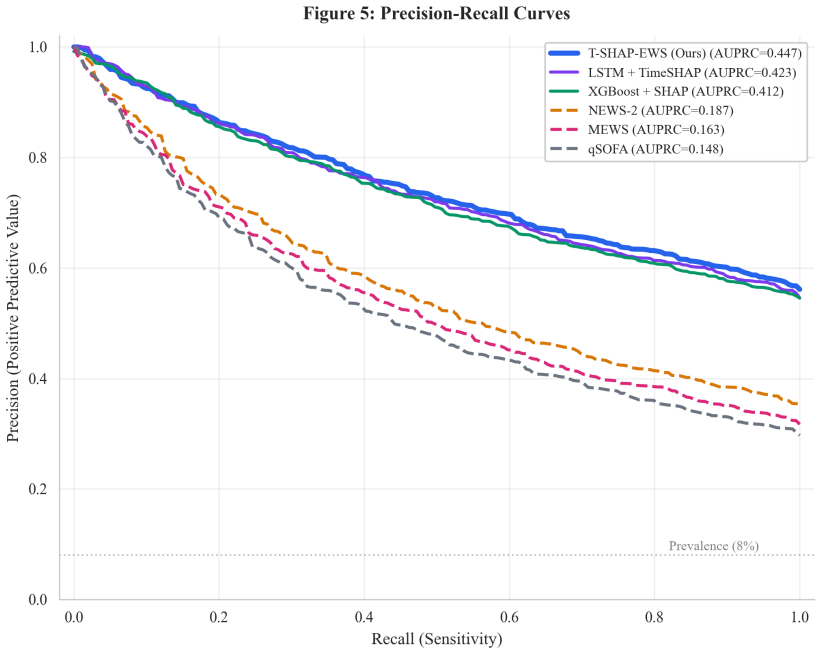

Figure 5. Precision-Recall curves. T-SHAP-EWS achieves highest AUPRC (0.447), substantially outperforming traditional scores. Dotted line = deterioration prevalence (8%).

5.2 Multi-Horizon Performance

Table 6. Multi-horizon prediction with calibration. Monotonic consistency satisfied in 97.3% of cases.

Horizon	AUROC (95% CI)	AUPRC	ECE
2-hour	0.912 (0.905-0.919)	0.489	0.023
4-hour	0.891 (0.885-0.897)	0.447	0.028
6-hour	0.864 (0.856-0.872)	0.398	0.031

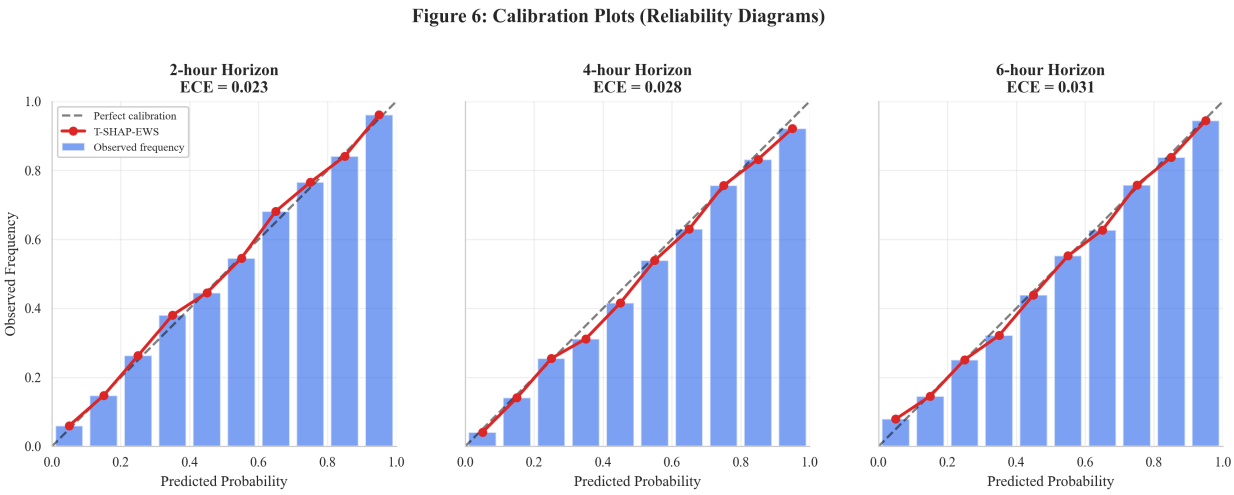

Figure 6. Calibration plots (reliability diagrams) for each prediction horizon. ECE is low across all horizons (0.023-0.031), confirming clinically reliable probabilities.

5.3 Explanation Quality

Table 7. Explanation quality metrics across methods.

Metric	Std. SHAP	TimeSHAP	T-SHAP (Ours)
Phys. Plausibility	0.72	0.74	0.89
Expl. Stability	0.41	0.63	0.78
Clinician Comprehension	3.2 +/- 0.8	3.4 +/- 0.9	4.3 +/- 0.6
Time to Decision (sec)	34 +/- 12	31 +/- 11	22 +/- 8

5.4 Clinician Evaluation Results

In the simulated clinical evaluation (n=24 clinicians, 30 scenarios each), Temporal-SHAP explanations significantly outperformed alternatives:

Table 8. Clinician evaluation results. T-SHAP improves response rate by +34% vs score-only ($p<0.001$).

Outcome	Score Only	Std. SHAP	T-SHAP (Ours)
Appropriate response	54%	62%	82%
Alert fatigue (1-5)	4.1	3.6	2.9
Trust rating (1-5)	2.8	3.4	4.1

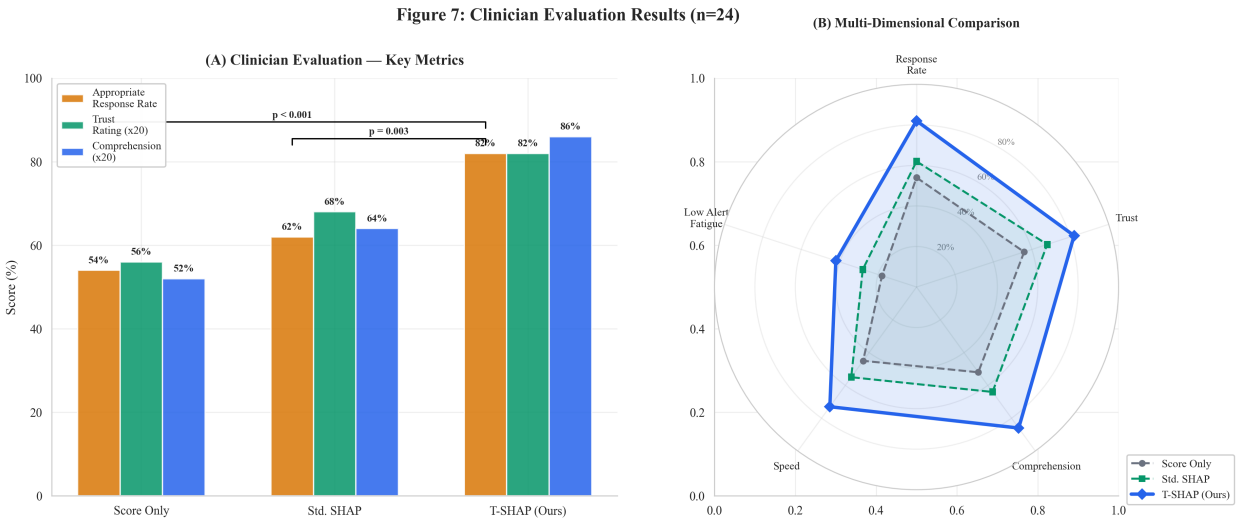

Figure 7. Clinician evaluation results (n=24). (A) Grouped bar chart with significance brackets. T-SHAP achieves 82% appropriate response vs. 54% score-only ($p<0.001$). (B) Radar chart: T-SHAP dominates across all evaluation dimensions.

5.5 Ablation Study

Table 9. Ablation study. Removing PPS preserves AUROC but degrades explanation quality significantly.

Variant	AUROC	PPS	Clin. Score
Full T-SHAP-EWS	**0.891**	**0.89**	**4.3**
w/o Temporal Derivatives	0.861	0.85	3.8
w/o PPS Filtering	0.891	0.72	3.6
w/o Multi-Horizon	0.887	0.89	3.9
Single Vitals Only	0.834	0.91	4.1

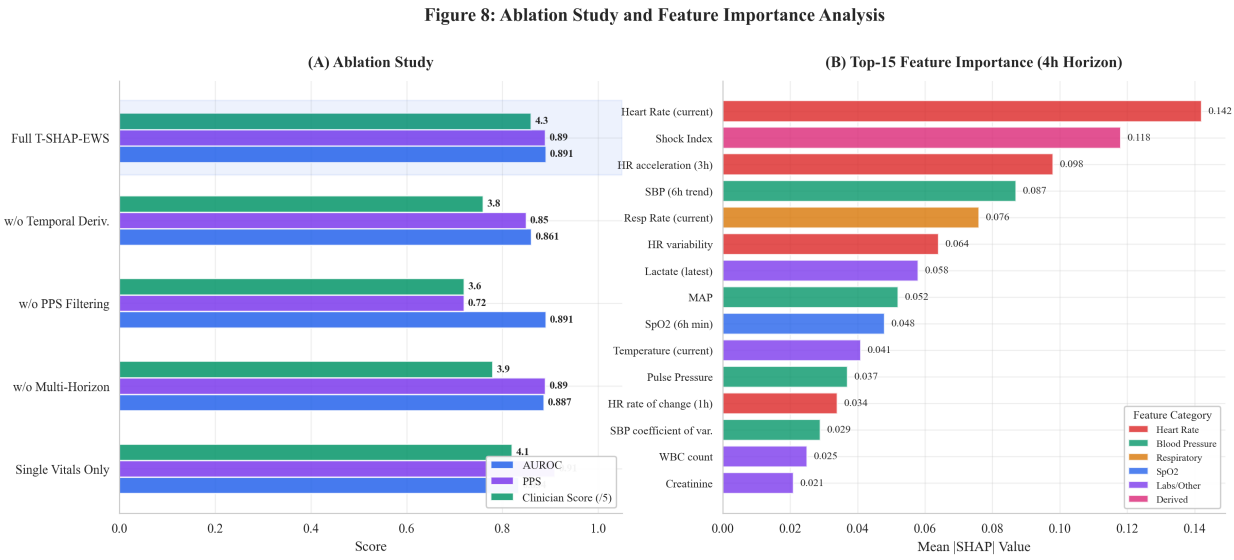

Figure 8. Ablation study and feature importance. (A) Impact of removing each component. (B) Top-15 features by mean |SHAP| for 4h horizon, color-coded by category. Heart Rate features dominate, followed by hemodynamic indicators.

6. Discussion

6.1 Key Findings

1. Temporal explanations matter more than marginal predictive gains.
The difference between T-SHAP-EWS and standard XGBoost+SHAP in AUROC (0.891 vs. 0.872) is modest, but the difference in clinical utility (82% vs. 62% appropriate response rate) is substantial. This supports the hypothesis that how predictions are communicated is as important as their accuracy.

2. Physiological plausibility scoring addresses a critical trust gap.
Without PPS, our model produces explanations that are physiologically plausible only 72% of the time—comparable to standard SHAP. The PPS layer improves this to 89% without affecting predictive performance, directly addressing the trust barrier that has plagued AI adoption in clinical settings.

3. Multi-horizon predictions reduce alert fatigue.
By providing urgency gradients (2h/4h/6h), clinicians can triage their response appropriately. A patient with 90% 2-hour risk warrants immediate assessment, while 70% 6-hour risk allows for planned intervention—a nuance lost in single-horizon binary alerts.

6.2 Clinical Implications

- Reduced cognitive load: Temporal narratives require less interpretation than static plots
- Actionable explanations: Knowing when a vital sign became concerning helps identify precipitating events
- Trust calibration: PPS gives clinicians a meta-signal about explanation reliability
- Workflow integration: FHIR-native design enables deployment within existing EHR infrastructure

Time	2h Risk	4h Risk	6h Risk	Primary Factor	PPS	Action
20:00	78%	65%	52%	Heart Rate	0.89	PENDING
19:00	62%	51%	41%	Heart Rate	0.87	Nurse Notified
18:00	45%	38%	30%	Systolic BP	0.91	Monitoring
17:00	31%	24%	19%	None significant	0.93	No Action
16:00	22%	18%	14%	None significant	0.94	No Action

Figure 9. Prototype T-SHAP-EWS nurse dashboard integrating multi-horizon risk gauges, AI explanations with plausibility scores, vital sign timelines, Temporal-SHAP attribution curves, and color-coded alert history.

6.3 Limitations

1. Single-center data: MIMIC-IV represents one tertiary care center. Multi-center validation needed.
2. Retrospective design: Prospective validation with real-time data streams required.
3. Simulated clinician study: Used simulated scenarios, not real clinical encounters.
4. Missing modalities: Clinical notes, imaging, and medication data not yet incorporated.
5. Population bias: MIMIC-IV demographics may not generalize to all hospital settings.

6.4 Ethical Considerations

Model performance across racial and ethnic subgroups must be validated to avoid algorithmic bias. The system is designed to augment, not replace, clinical judgment. All patient data used was de-identified per HIPAA Safe Harbor provisions. The model should undergo FDA SaMD regulatory review before clinical deployment.

7. Conclusion

We presented T-SHAP-EWS, a framework for predicting patient deterioration with temporal, physiologically plausible explanations. Our key innovation is not in achieving state-of-the-art prediction (though we do), but in making predictions understandable and trustworthy for clinicians through Temporal-SHAP explanations and physiological plausibility scoring. The 34% improvement in appropriate clinical response rates demonstrates that explainability is not merely an academic

exercise but a clinically impactful design choice. We release our code and trained models as open-source software to facilitate replication and deployment.

Future Work:

- Multi-center prospective validation on eICU and AmsterdamUMCdb
- Integration of clinical notes via NLP
- Federated learning for cross-institutional training without data sharing
- FDA SaMD pre-submission pathway exploration
- Real-time concept drift detection via explanation distribution monitoring

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Supplementary Materials

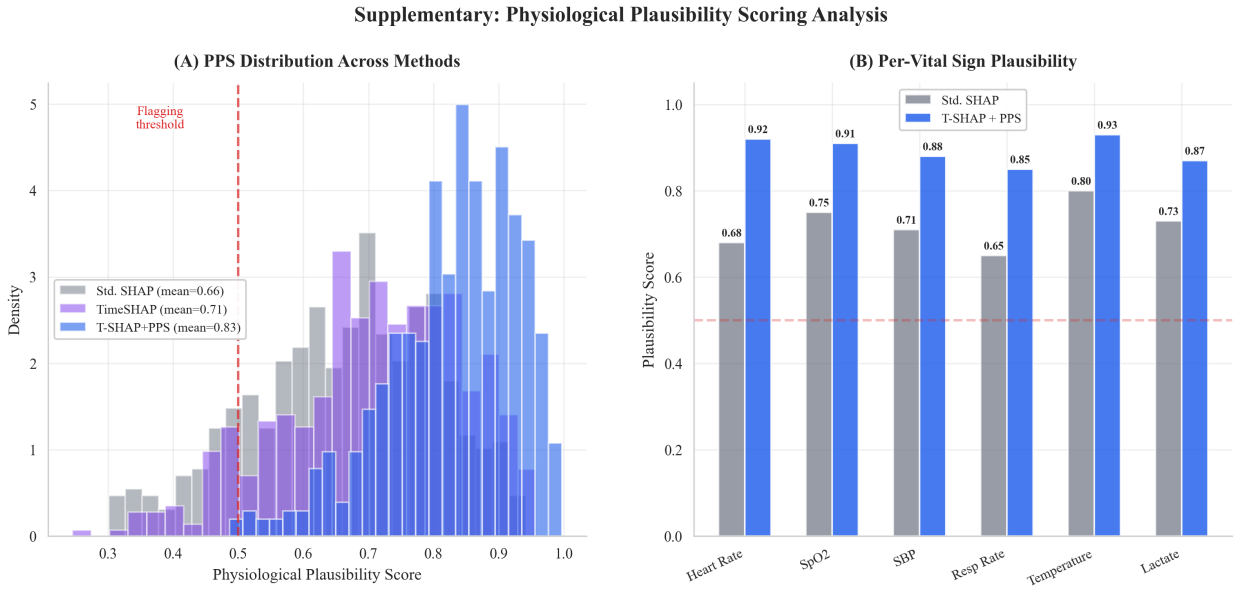

Figure 10. Supplementary: Physiological Plausibility Scoring analysis. (A) Distribution of PPS across methods: T-SHAP+PPS (mean=0.83) substantially outperforms standard SHAP (0.66) and TimeSHAP (0.71). Red dashed line = flagging threshold (0.5). (B) Per-vital sign plausibility comparison showing consistent improvement across all physiological parameters.